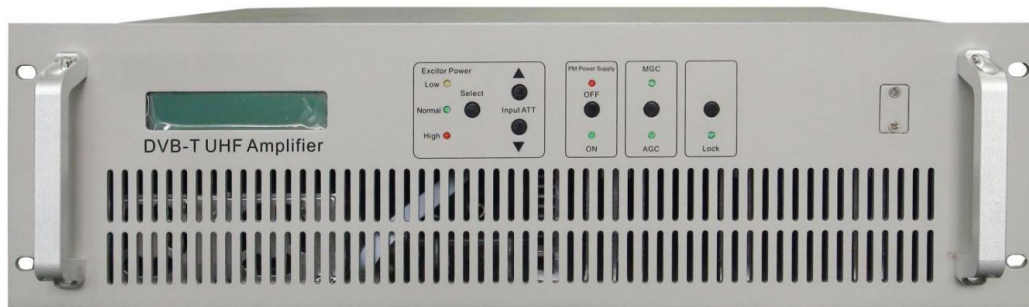


## FMUSER DUT-8351 Digital Transmitter 1-Channel DVB-T-ATSC-ISDB-T Input Output Adjustable 10-100W RF Power LDMOS Amplifier Indoor HD SD Broadcasting Product overview



The FUQ-8351 is a professional indoor digital transmitter supporting DVB-T/ATSC/ISDB-T standards with 10-100W adjustable RF output (470-806MHz). It processes signals via 4/16/32/64 QAM modulation and features AGC-controlled power stability ( $\pm 0.25\text{dB}$ ). The 2U rack-mount unit includes LED alarms for real-time fault monitoring, designed for SFN/MFN networks in broadcast studios and mobile TV systems.

### Key feature

Broadcast-Grade RF Performance: Achieves shoulder  $\geq 40\text{dB}$  across 470-806MHz frequency range

Monitoring Objects and Alarm Status

| Monitoring Objects  | Alarm Status |
|---------------------|--------------|
| Power Supply        | abnormal     |
| Overheating         | alert        |
| Amplifier           | abnormal     |
| Antenna feed system | abnormal     |

### Phase noise

| Center frequency deviation (Hz) | Local oscillator (dBc/Hz) |
|---------------------------------|---------------------------|
| @10                             | $\leq -70$                |
| @100                            | $\leq -85$                |
| @1k                             | $\leq -95$                |

|        |             |
|--------|-------------|
| @10 k  | $\leq -100$ |
| @100 k | $\leq -111$ |
| @1 M   | $\leq -125$ |

#### Electrical Parameters

| Item             | Item                                         | Technical specs                                                                                    |
|------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| Basic Parameters | TV Standard                                  | DVB-T/ATSC/ISDB-T                                                                                  |
| Basic Parameters | Modulating mode                              | 4/16/32/64 QAM                                                                                     |
| Basic Parameters | Frequency stability (3 months)               | External Frequency Range:<br>$\leq 10^{-10}$                                                       |
| Basic Parameters | Frequency stability (3 months)               | Internal Frequency Range:<br>$\leq 10^{-7}$                                                        |
| Basic Parameters | Frequency accuracy                           | MFN: $\pm 100\text{Hz}$ / SFN: $\pm 1\text{Hz}$                                                    |
| Basic Parameters | Local oscillator phase noise                 | See table above                                                                                    |
| Basic Parameters | In-band stray                                | $\leq -60\text{dBc}$                                                                               |
| Basic Parameters | In-band ripple $f_c \pm 3.591\text{MHz}$     | $\pm 0.5\text{dB}$                                                                                 |
| Basic Parameters | Out-of-band rejection                        | $\leq -65\text{dBc}$                                                                               |
| Input            | Frequency range                              | 470MHz~806MHz (any 100MHz bandwidth)                                                               |
| Input            | level                                        | $-20 \pm 3\text{dBm}$                                                                              |
| Input            | Input reflection loss                        | $\geq 15\text{dB}$                                                                                 |
| Input            | connector                                    | N-K                                                                                                |
| Output           | Output power                                 | 10W, 20W, 30W, 50W, 100W<br>Optional                                                               |
| Output           | Frequency range                              | 470MHz~806MHz (any 100MHz bandwidth)                                                               |
| Output           | Output impedance                             | 50 $\Omega$                                                                                        |
| Output           | In-band ripple ( $f_c \pm 3.591\text{MHz}$ ) | $\pm 0.5\text{dB}$                                                                                 |
| Output           | Shoulder level                               | $\geq 40\text{dB}$ @central frequency<br>FC $\pm 4.2\text{MHz}$ (after correction, single channel) |

|                       |                                              |                                                                                                                                                                        |
|-----------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output                | MER                                          | $\geq 35\text{dB}$ (after correction, single channel)                                                                                                                  |
| Output                | output reflection loss                       | $\geq 26\text{dB}$                                                                                                                                                     |
| Output                | Output power variation                       | $\pm 0.25\text{dB}$                                                                                                                                                    |
| Output                | Inner adjacent channel transmission power    | Inner adjacent channel transmission power : in-band transmission power $\leq -60\text{dB}$ , meet that Inner adjacent channel transmission power $\leq 13\text{mW}$    |
| Output                | External adjacent channel transmission power | External adjacent channel transmission power : in-band transmission power $\leq -65\text{dB}$ , meet that Inner adjacent channel transmission power $\leq 13\text{mW}$ |
| Environment condition | Working temperature                          | $-20\sim+50^{\circ}\text{C}$                                                                                                                                           |
| Environment condition | Storage temperature                          | $-30\sim+75^{\circ}\text{C}$                                                                                                                                           |
| Environment condition | Relatively humidity                          | $\leq 95\%$ no condensation at $25^{\circ}\text{C}$                                                                                                                    |
| Environment condition | atm press                                    | $86\sim 106\text{kPa}$                                                                                                                                                 |
| Environment condition | power supply                                 | AC, $220\text{V}\pm 10\%/50\text{Hz}$                                                                                                                                  |
| Environment condition | Case size (L/W/H)                            | $482\times 504\times 132\text{mm}$                                                                                                                                     |

#### Optical Signal Parameters

| Item                        | Unit | Technical Specs |
|-----------------------------|------|-----------------|
| Optical wavelength          | nm   | 1100-1600       |
| nominal optical input power | dBm  | -1              |
| Input optical power range   | dBm  | $-15\sim+2$     |
| Max input optical power     | dBm  | +3              |
| Optical interface           |      | FC/APC          |
| optical reflection loss     | dB   | $\leq 45$       |

## About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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